PD Controller with Dynamics Compensation

LVLH Tracking and Lyapunov Stability Proof

1. Model and errors

Consider the current quaternion controller. Let I be the inertial frame, B the spacecraft body frame, and D the desired LVLH frame. For nadir pointing, the desired $+z_D$ axis points towards the centre of Earth.

Euler's equation is

$$J\dot{\omega} + \omega \times (J\omega) = \tau + \tau_{\text{ext}}, \qquad J = J^T > 0, \qquad \omega = \omega_{BI}^B.$$

Here J is the constant inertia matrix, τ is the control torque, and τ_{ext} is the disturbance torque. In the proof we assume $\tau_{\text{ext}} = 0$, exact knowledge of the state and J, continuous and active control, and no torque saturation.

Let C_{BI} map vector components from I to B, and C_{DI} from I to D. Then $C_{BD} = C_{BI}C_{DI}^T$ maps components from D to B. The corresponding error quaternion has four scalar components:

$$q_e = \begin{bmatrix} \eta \\ \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix}, \qquad \eta^2 + \varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 = 1.$$

We use the scalar-first convention of the model blocks. In the implementation,

$$q_e = \text{quatmultiply}(\text{quatconj}(q_{DI}), q_{BI}).$$

The sign of all four components is chosen so that $\eta \geq 0$. The proof stays within $\eta > 0$. The desired error quaternion is $q_e = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}^T$; the reference quaternion $q_{DI}(t)$ itself need not be constant.

For vector operations, define the three-component vector $\varepsilon = \begin{bmatrix} \varepsilon_1 & \varepsilon_2 & \varepsilon_3 \end{bmatrix}^T$. This is the vector part of q_e, not a separate quaternion. Let

$$\omega_d = \omega_{DI}^D, \qquad \alpha_d = \frac{d\omega_{DI}^D}{dt}, \qquad r = C_{BD}\omega_d, \qquad a = C_{BD}\alpha_d.$$

Here α_d is the derivative of the reference components expressed in D. We assume that the reference attitude, rate and acceleration are kinematically consistent. The rate error is expressed entirely in B:

$$\boxed{e_\omega = \omega - r.}$$

We do not subtract vectors expressed in different frames.

2. Compensation and control law

The term $\omega \times (J\omega)$ couples the axes. Let

$$\tau = \omega \times (J\omega) + Jv,$$

where v is a virtual input. Euler's equation then gives $\dot{\omega} = v$. For the rotating reference, we also have

$$\dot{C}_{BD} = -[e_\omega]_\times C_{BD}, \qquad \dot{r} = -e_\omega \times r + a, \qquad [u]_\times w = u \times w.$$

Therefore,

$$\dot{e_\omega} = \dot{\omega} - \dot{r} = \dot{\omega} + e_\omega \times r - a.$$

Choose

$$v = -e_\omega \times r + a - 2k_p\varepsilon - k_d e_\omega, \qquad k_p > 0, \quad k_d > 0.$$

Thus the control law is

$$\boxed{\tau = \omega \times (J\omega) + J(-e_\omega \times r + a - 2k_p\varepsilon - k_d e_\omega).}$$

Substitution gives the exact error dynamics

$$\boxed{\dot{e_\omega} = -2k_p\varepsilon - k_d e_\omega.}$$

The gains are scalar and equal on all axes: $K_p = k_p I_3$, $K_d = k_d I_3$. Their units are $[k_p] = \mathrm{s}^{-2}$, $[k_d] = \mathrm{s}^{-1}$; $[\tau] = \mathrm{N\,m}$ and $[J] = \mathrm{kg\,m}^2$.

Why the factor of 2? The attitude error signal in the current model is 2ε. For an error angle θ about a unit axis n, we have

$$\varepsilon = n \sin\frac{\theta}{2}, \qquad 2\varepsilon \simeq n\theta \quad \text{for small angles.}$$

Hence the term $-2k_p\varepsilon$. Writing $-k_p\varepsilon$ instead corresponds to a different gain scaling.

What has been linearised? The compensation cancels the nonlinear terms in the rate dynamics. The quaternion kinematics remain nonlinear. Writing $e_\omega = \begin{bmatrix} e_1 & e_2 & e_3 \end{bmatrix}^T$, they are

$$\begin{bmatrix} \dot{\eta} \\ \dot{\varepsilon}_1 \\ \dot{\varepsilon}_2 \\ \dot{\varepsilon}_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -\varepsilon_1 & -\varepsilon_2 & -\varepsilon_3 \\ \eta & -\varepsilon_3 & \varepsilon_2 \\ \varepsilon_3 & \eta & -\varepsilon_1 \\ -\varepsilon_2 & \varepsilon_1 & \eta \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}.$$

Only for a small angle-error vector $\theta_e \simeq 2\varepsilon$ do we obtain $\dot{\theta}_e \simeq e_\omega$, and therefore

$$\ddot{\theta}_e + k_d\dot{\theta}_e + k_p\theta_e \simeq 0.$$

This is a local second-order approximation on each axis. The proof below uses the exact quaternion kinematics.

3. Lyapunov function and invariant set

Choose

$$\boxed{V(q_e, e_\omega) = 4k_p(1-\eta) + \frac{1}{2}e_\omega^T e_\omega.}$$

The first term describes the attitude error and the second the rate error. The units of V are s^{-2}; it is not the physical rotational energy. On the unit quaternion sphere, $\eta \leq 1$. Therefore,

$$V = 0 \iff \eta = 1, \quad \varepsilon_1 = \varepsilon_2 = \varepsilon_3 = 0, \quad e_\omega = 0,$$

and $V > 0$ everywhere else.

From the first row of the quaternion kinematics,

$$\dot{\eta} = -\frac{1}{2}(\varepsilon_1 e_1 + \varepsilon_2 e_2 + \varepsilon_3 e_3) = -\frac{1}{2}\varepsilon^T e_\omega.$$

The derivative along trajectories is

$$\begin{aligned}\dot{V} &= -4k_p\dot{\eta} + e_\omega^T \dot{e_\omega} \\ &= 2k_p\varepsilon^T e_\omega + e_\omega^T(-2k_p\varepsilon - k_d e_\omega) \\ &= -k_d e_\omega^T e_\omega \leq 0,\end{aligned}$$

since $\varepsilon^T e_\omega = e_\omega^T \varepsilon$. The derivative is negative semidefinite. This gives stability, but the inequality alone does not yet establish convergence.

Let $x = (q_e, e_\omega)$ and define

$$\Omega_c = \{(q_e, e_\omega) \in S^3 \times \mathbb{R}^3 : V(q_e, e_\omega) \leq c\}, \qquad 0 < c < 4k_p,$$

where S^3 is the unit quaternion sphere. Since $\dot{V} \leq 0$,

$$x(0) \in \Omega_c \implies V(x(t)) \leq V(x(0)) \leq c \implies x(t) \in \Omega_c \qquad (t \geq 0).$$

Thus Ω_c is positively invariant. It is also compact: it is closed, the quaternion has unit norm, and $\|e_\omega\|^2 \leq 2c$.

Why $c < 4k_p$? Since the second term of V is nonnegative,

$$4k_p(1-\eta) \leq V \leq c < 4k_p \quad \implies \quad \eta \geq 1 - \frac{c}{4k_p} > 0.$$

The trajectory therefore stays on one smooth quaternion branch, without reaching an attitude error of 180°. A quaternion still exists at 180°; the issue is the sign choice at the branch boundary, not a quaternion singularity.

4. LaSalle's invariance principle

Let

$$E = \{x \in \Omega_c : \dot{V}(x) = 0\} = \{(q_e, e_\omega) \in \Omega_c : e_\omega = 0\}.$$

We find the largest invariant set contained in E. For $e_\omega = 0$, the dynamics give

$$\dot{e_\omega} = -2k_p \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix}.$$

To remain in E, we need $\dot{e_\omega} = 0$, so $\varepsilon_1 = \varepsilon_2 = \varepsilon_3 = 0$. The unit quaternion condition gives

$$\eta^2 + \varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 = 1 \quad \Longrightarrow \quad \eta = \pm 1.$$

But $\eta > 0$ in Ω_c, hence $\eta = 1$. At this point, $\dot{q}_e = 0$ as well. The largest invariant set contained in E is the single point

$$M = \left\{ (q_e, e_\omega) = \left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right) \right\}.$$

After exact compensation, the error system is autonomous. By LaSalle's invariance principle, every trajectory starting in Ω_c satisfies

$$q_e(t) \longrightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad e_\omega(t) \longrightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad (t \to \infty).$$

Positive definiteness of V and $\dot{V} \leq 0$ give stability, and the above convergence gives attraction. Hence the equilibrium is **locally asymptotically stable in the sense of Lyapunov**.

5. Scope of the proof

A sufficient initial condition is $V(0) < 4k_p$. For $e_\omega(0) = 0$, this includes every attitude error below $180°$. It does not describe the entire region of attraction and is not a global proof.

The proof concerns the ideal model. The simulation uses MEKF estimates, sampled control, torque saturation and a quality gate. The reference rate is approximate, and acceleration is obtained by a backward difference. These effects are not covered by the proof.

For example, with a nonzero disturbance torque and the other ideal assumptions,

$$\dot{e_\omega} = -2k_p\varepsilon - k_d e_\omega + J^{-1}\tau_{\text{ext}}, \qquad \dot{V} = -k_d\|e_\omega\|^2 + e_\omega^T J^{-1}\tau_{\text{ext}}.$$

We can no longer guarantee $\dot{V} \leq 0$. The current PD controller may therefore leave a steady attitude error. Disturbance torque compensation requires a separate extension of the controller and proof.

6. How close to the desired attitude must we start?

Let $\theta_0 \in [0, \pi]$ be the initial attitude error angle. Since $\eta(0) = \cos(\theta_0/2)$, the condition $V(0) < 4k_p$ gives

$$4k_p \left(1 - \cos\frac{\theta_0}{2}\right) + \frac{1}{2}\|e_\omega(0)\|^2 < 4k_p.$$

Rearranging, we obtain

$$\boxed{\|e_\omega(0)\| < \sqrt{8k_p \cos\frac{\theta_0}{2}}.}$$

Thus, the larger the initial attitude error, the smaller the rate error allowed by this proof. We use the norm of the entire vector $e_\omega(0)$, not each component separately.

For $k_p = 0.0004$ s^{-2}, the bounds are:

Angle θ_0	Bound on $\|e_\omega(0)\|$	
	[rad/s]	[°/s]
0°	0.05657	3.241
30°	0.05560	3.185
60°	0.05264	3.016
90°	0.04757	2.725
120°	0.04000	2.292
150°	0.02878	1.649
170°	0.01670	0.957
175°	0.01181	0.677
179°	0.00528	0.303

The table entries are rounded. The error norm must be **strictly smaller** than the bound computed from the formula. Here e_ω is the error relative to the LVLH tracking rate, not the spacecraft angular velocity itself.

We can also find the angle bound for a given rate error:

$$\boxed{\theta_0 < 2\arccos\left(\frac{\|e_\omega(0)\|^2}{8k_p}\right),} \qquad \|e_\omega(0)\| < \sqrt{8k_p}.$$

For example, for $\|e_\omega(0)\| = 2°/\text{s}$, after converting the rate to rad/s, we obtain $\theta_0 < 135.24°$ (rounded bound). For $e_\omega(0) = 0$, the condition reduces to $\theta_0 < 180°$.

Thus local stability does not mean only small angles here. The condition is sufficient, but not necessary. Outside this region, the controller may still converge; this proof does not decide that. The table uses the same ideal assumptions as the proof, in particular the absence of torque saturation.